First-birth housing: a measurement issue to resolve

Calibration assessment for the September 14 presentation

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The decision for discussion

Resolve the empirical housing statistic before another calibration search. The audit identifies a normalization problem in the event-study design: some birth cohorts have no observation in the stated reference period, and their weights differ between the two endpoints used to construct the target. An arbitrary shift in their coefficient levels can therefore change the reported contrast without changing fitted housing outcomes. An independent read-only review confirms the algebra. The exact omitted coefficients in the production regression still need to be recovered.

This finding does **not** supply a corrected rooms target, establish the size or sign of its error, or explain the whole model shortfall. The recorded target remains **0.720246 rooms**, with its existing reported standard error **0.085260**. The existing model, target file, weights and calibration results are unchanged. The appendix retains every target-fit row and estimated parameter for both the production calibration and the verified refinement candidate.

Why the reference period matters

A birth cohort is the group of women with a first birth in the same calendar year. The estimator fits a separate housing profile for each cohort and then averages those profiles using the cohort shares observed at each event time. The reference period, two years before first birth, is assigned a zero effect.

The sample-only replay finds **16 birth cohorts with no observation at event year -2**. Such cohorts account for **41.91%** of the cohort-share weight at -1 and **46.01%** at +3. A second sample-only check verifies that all regression inputs are nonmissing, first-birth cohort is constant within person, and the included event indicators cover every treated observation except -2. These checks require no regression coefficients.

For a cohort without the reference observation, shifting all its observed housing coefficients up by one room and its person fixed effects down by one room leaves every fitted observation unchanged. If that cohort receives different weights at the two endpoints, the weighted difference changes. That is a failure of invariance to an arbitrary normalization, beyond the usual question of whether changing cohort composition is economically desirable.

A concrete check of the normalization

Let $\delta_{g,k}$ denote the cohort-g housing coefficient at event time k, and $w_{g,k}$ its cohort share at that event time. The reported target is

$$D = \sum_g w_{g,3}\delta_{g,3} - \sum_g w_{g,-1}\delta_{g,-1}.$$

For a cohort with no -2 observation, the transformation $\delta_{g,k} \mapsto \delta_{g,k} + c_g$ on its supported event cells and $\alpha_i \mapsto \alpha_i - c_g$ for its members' person fixed effects leaves the regression's predictions unchanged. The reported contrast, however, changes by

$$\Delta D = (w_{g,3} - w_{g,-1})c_g.$$

The **1986 first-birth cohort** supplies a numerical example with observations at both endpoints. It has no -2 observation; its weights are **0.026299** at -1 and **0.040435** at +3. A one-room normalization shift changes the contrast by **0.014136 rooms**, while the verified change in every fitted row is **zero**. The one-room shift is an algebraic illustration, not an economic intervention, an estimate of bias, or a proposed correction.

The installed estimator calculates these cohort shares before the regression's singleton pruning. The all-inputs-observed assertion verifies that the saved input support is the sample used for that share calculation. The zero-prediction identity also holds on any subset retained by the regression. What remains unrecovered is the realized estimation sample, the omitted-column choices and the cohort coefficient/covariance objects needed to quantify an alternative well-defined contrast. The original receipt reports 49,457 estimation rows; the verified input sample contains 49,872 rows.

A common set of cohort weights applied to estimable within-cohort changes would cancel this particular arbitrary level. That is a candidate diagnostic repair, not an approved replacement target: cohort overlap, remaining rank conditions, uncertainty and economic interpretation must all be checked. No target is dropped, demoted or reweighted in this review.

What passed, and what did not

The source and target provenance verification passed, including the raw PSID source hash. The first sample-only replay passed in **28 seconds**, and the additional design assertions passed in **84 seconds**. The one predeclared full regression replay reached its **900-second limit** and was stopped; its recorded elapsed time is 902.5 seconds including termination. It produced no usable cohort coefficients or covariance matrix. No second regression was launched. Post-run hashes confirm that the primary builder and every primary target output remain unchanged. Only cohort/event aggregates were exported.

The event-time pattern does not supply a convenient replacement

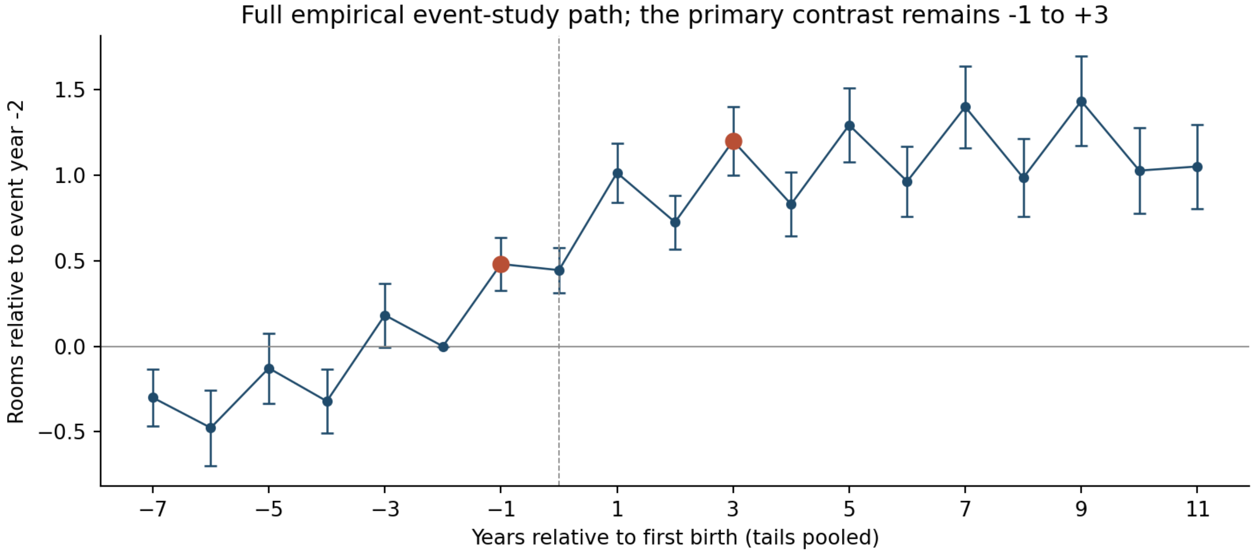

Figure 1: Supplemental empirical profile from the unchanged production event table. Bars show its existing pointwise 95 percent intervals, conditional on the reported specification and normalization.

The profile alternates substantially between odd and even event years. However, nearby four-year summaries of the saved coefficients remain near 0.72 rooms:

Contrast	Rooms	Interpretation
-1 to +3	0.720246	Existing target; reported SE 0.085260
-2 to +2	0.726160	Diagnostic; reported SE 0.080315
Mean of +2/+3 minus mean of -2/-1	0.723203	Diagnostic; joint uncertainty unavailable
0 to +4	0.385433	Different object, excluding adjustment through the birth year

These comparisons are arithmetic diagnostics from the old coefficient table, not newly estimated or validated causal effects. The missing covariance prevents an uncertainty calculation for their differences. In particular, the smaller 0-to-4 value is not a reason to replace the target with a number closer to the model.

The treated input observations span **1970-2019**. The weighted mean first-birth year is **1994.64** at -1 and **1992.97** at +3; the share of weight on birth cohorts from 2007 onward falls from **21.39% to 18.05%**. The total variation distance between endpoint cohort shares is **0.20236**: 20.24 percentage points of cohort probability weight would have to be reassigned to make the two mixes identical. These are cohort-share input statistics, not claims about the final regression sample or a quantified correction to the target.

What the model measures

The active housing statistic compares two copies of the households that successfully have a first birth. One copy has the baby; the other remains childless. Both begin with the same wealth, age, income, tenure and location, and follow their respective saving and housing choices over one four-year model period. The difference in their destination rooms is also a difference in changes because their starting states are identical.

At the review candidate, the 2019-2023 treated copy occupies **5.800342 rooms** and the childless copy **5.334298**, giving **0.466044 rooms**. The treated branch already permits a later birth at the destination date: continuation births equal **26.88%** of its mass. Therefore, a missing second-birth option in this active measurement is not the explanation for the shortfall. An older same-policy helper has different timing and must not be confused with this dated calculation.

Comparison	Empirical target	Model statistic
Population	PSID women who are reference persons or spouses, with one observation per household-year	Successful first-birth households in the fitted first-birth risk set
Control	Women confirmed childless in their observed histories	Identical copies held childless through the window
Clock	First-birth event years -1 and +3	One four-year interval, 2019-2023 for the fitted terminal moment
Calendar coverage	Pooled historical PSID interviews	The fitted terminal four-year cohort
Before-birth differences	Estimated from an observational panel; anticipation and selection remain relevant	Starting-state differences between copies are zero by construction
Later births	Allowed in treated mothers' histories	Allowed in the treated branch at the destination

Matching the duration does not establish that these are identical economic objects. In particular, the model comparison holds the initial state fixed for households selected into first birth, whereas the empirical estimator uses a separate childless population. Neither exact reproduction nor a small standard error establishes the missing comparability assumptions. No new estimate of the size of this model-to-data selection discrepancy is available from this replay.

What this means for calibration

The best review candidate reduces the unchanged twelve-target loss from **30.482967 to 23.153445**. Across the **35 coordinate and joint evaluations**, the largest first-birth response is **0.466044 rooms** and the smallest average housing level is **6.270186 rooms**. Those extrema do not occur at the same parameter vector. The target pair is **0.720246** and **5.779970 rooms**. This neighborhood contains a meaningful improvement but provides no evidence that the two housing discrepancies have been jointly resolved. It is not a proof that the target pair is globally unreachable.

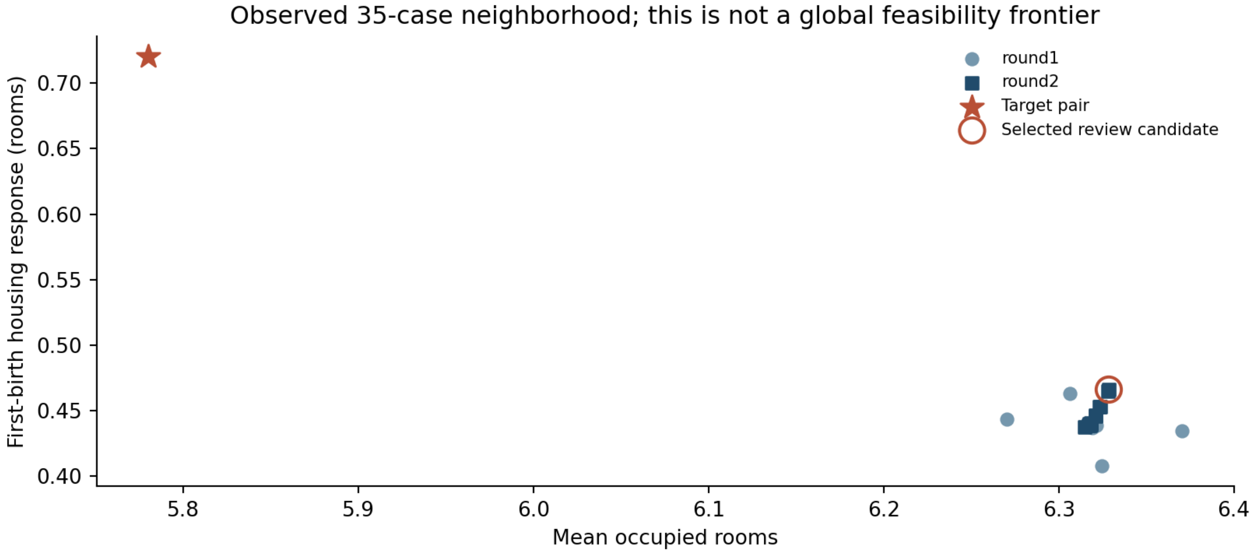

Figure 2: Supplemental housing-fit comparison using saved evaluations. No new model solution is computed.

After resolving the empirical normalization, the model measurement exercise should reproduce a stated empirical population and event clock in the model, including before-birth housing choices, and compare that statistic with the current matched-copy response. It should preserve the full twelve-moment fit table and distinguish a measurement change from an improvement under the existing objective. A new empirical weighting rule requires its own uncertainty calculation and target-contract decision.

Until that comparison is resolved, retain both the production calibration and the verified refinement candidate for discussion. The principal question is how much of the rooms shortfall reflects housing-demand parameters, and how much reflects comparing different populations and timing conventions. Another broad search alone cannot answer it. Existing policy numbers belong to the retained production calibration; this report supplies no revised policy or welfare result.

Complete target fit: verified candidate

Gaps equal model minus target. Contributions are weight times squared gap. The weights retain their existing empirical or synthetic interpretations; this loss is not a specification-test statistic.

Moment	Target	Model	Gap	Weight	Loss
Completed fertility	1.918000	1.921698	+0.003698	1425.74	0.0195
Childlessness	0.188000	0.189859	+0.001859	17180.7	0.0594
Mean age at first birth	26.044627	26.251703	+0.207076	44.4444	1.9058
First births at age 30+	0.260327	0.237163	-0.023165	10000	5.3660
Rooms response to first birth	0.720246	0.466044	-0.254202	137.565	8.8893
Rooms, 3+ minus 1-2 children	0.367700	0.379198	+0.011499	2958.51	0.3912
Family ownership gap	0.167662	0.169578	+0.001917	14229.6	0.0523
Ownership, ages 30-55	0.575472	0.546618	-0.028854	1207.85	1.0056
Mean occupied rooms	5.779970	6.328594	+0.548624	11.9732	3.6038
Wealth / annual earnings	6.873100	7.031196	+0.158096	6.28767	0.1572
Bequest flow / wealth	0.008800	0.008485	-0.000315	5.16529e+06	0.5139
Older wealth/income p90 / p50	3.448111	3.303595	-0.144516	56.9598	1.1896

Complete target fit: retained production

Moment	Target	Model	Gap	Weight	Loss
Completed fertility	1.918000	1.922907	+0.004907	1425.74	0.0343
Childlessness	0.188000	0.189407	+0.001407	17180.7	0.0340
Mean age at first birth	26.044627	26.256032	+0.211404	44.4444	1.9863
First births at age 30+	0.260327	0.237579	-0.022748	10000	5.1748
Rooms response to first birth	0.720246	0.436418	-0.283828	137.565	11.0821
Rooms, 3+ minus 1-2 children	0.367700	0.403441	+0.035741	2958.51	3.7793
Family ownership gap	0.167662	0.161699	-0.005963	14229.6	0.5060
Ownership, ages 30-55	0.575472	0.544488	-0.030984	1207.85	1.1596
Mean occupied rooms	5.779970	6.317291	+0.537321	11.9732	3.4568
Wealth / annual earnings	6.873100	6.932652	+0.059552	6.28767	0.0223
Bequest flow / wealth	0.008800	0.008433	-0.000367	5.16529e+06	0.6971
Older wealth/income p90 / p50	3.448111	3.236511	-0.211600	56.9598	2.5503

Complete parameter comparison

Parameter	Production	New candidate	Bounds / restriction	Near bound?
Annual patience β	0.995117	0.995739	[0.94, 0.9995]	No
First-birth dispersion κ_1	2.168173	2.168173	[0.02, 50]	No
Later-birth dispersion κ_C	1.736471	1.736471	[0.02, 50]	No
Owner-service premium χ	1.043472	1.043472	[0.1, 5]	No
Supply intercept H_0	14.562959	14.562959	[0.2, 80]	No
Bequest strength θ_0	0.528428	0.549189	[0, 8]	No
Bequest shifter θ_1	0.107249	0.107249	[0.02, 16]	Raw scale only
Per-child room floor	0.282210	0.248877	[0.1, 1.8]	No
First-birth fixed cost	4.559138	4.559138	[0, 8]	No
First-child room intercept	0.364931	0.464931	[0, 0.5]	No
2007-2023 child-value change	-0.328714	-0.328714	[-1.5, 0.2]	No
Tenure dispersion κ	0.005000	0.005000	Externally fixed	Not estimated
Dated supply elasticity η	0.630000	0.630000	Externally fixed	Not estimated

The old fertility-preference intercept is re-normalized to replacement for every parameter vector. New candidate values are $\psi_{2007} = 0.28947879$ and $\psi_{2023} = -0.03923512$. The stored near-bound flag uses raw parameter units. For the bequest shifter, interpret it alongside its position in the logarithmic search interval; a raw-scale flag alone does not establish a binding search constraint.

The supply restriction 0.63 applies to dated markets. The inherited initialization retains elasticity 1.75 before the dated supply intercept is normalized. Both values and that order are unchanged across this comparison; whether this is the intended economic contract remains a discussion item in the separate code audit.

Evidence and reproduction

All paths below are relative to the project root: `/Users/tommasodesanto/Desktop/Projects/Fertility/Fertility_Spring26`.

- Primary estimator: `code/data/psid_followup_mar2026/sa_rooms_first_birth_household_aligned_v1.do`, sample construction and event definitions at lines 65-147, estimator and contrast at lines 166-212.
- Primary target and original event table: `code/data/psid_followup_mar2026/output/sa_rooms_first_birth_household_aligned_v1/`.
- Sample replay driver: `code/data/psid_followup_mar2026/audit_first_birth_rooms_measurement.py`.
- Aggregate analysis and two supplemental figures: `code/model/tools/analyze_e5f_first_birth_measurement.py`.
- Complete receipts, aggregates, null-direction check, failed regression receipt and independent review: `output/model/e5f_first_birth_measurement_review_20260905a/`.
- Active matched-branch implementation: `code/model/tools/run_e5f_transition_calibration.py`, functions `begin_dated_first_birth_housing_branch` and `finish_dated_first_birth_housing_branch`.
- Saved candidate: `output/model/e5f_morning_refinement_20260905a/round2/task_009/`; its dated-branch CSV is under `cases/task_001/` within that folder. This inner case identifier does not change the selected outer candidate.
- Earlier complete calibration readout and standard seventeen-graph packet: `docs/model/e5f_bounded_calibration_refinement_review.md` and `output/pdf/e5f_bounded_calibration_refinement_review.pdf`.

The installed estimator was also inspected at `/Users/tommasodesanto/Library/ApplicationSupport/Stata/ado/plus/e/eventstudyinteract.ado`: cohort-share estimation at lines 40-47, interacted regression at line 85, and weighted aggregation at lines 103-108. The evidence manifest pins its hash.

Regenerate the aggregate analysis without a Stata regression or model solve:

```
python3 code/model/tools/analyze_e5f_first_birth_measurement.py
```

The output-folder README retains the original regression budget, both successful sample checks, the failed full replay and the PDF build command. Full per-case target and parameter tables are retained alongside this report. The two new figures are supplemental; the standard model diagnostic packet is unchanged.